

Lecture 6 – Normalization

CPE241 – Database Systems

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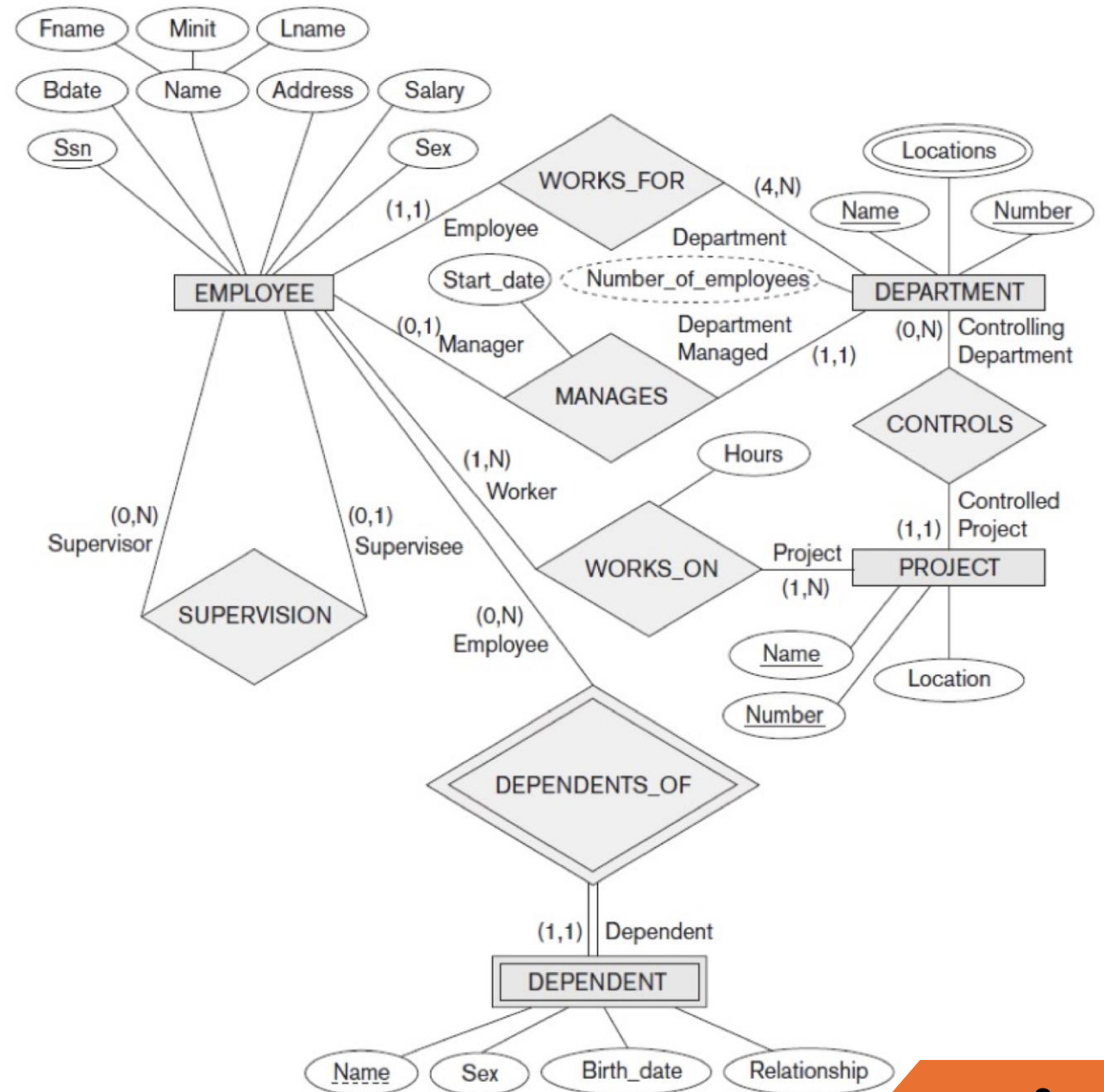
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Recap

- Real-world problem to an ERD:
COMPANY Database



Today's Goals

- An ERD to relational database
- Modification anomalies: insertion, deletion, update
- Functional dependency (FD)
- Normalization: 1NF, 2NF, 3NF, BCNF

Relational Database Schema

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
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DEPARTMENT

Dname	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
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DEPT_LOCATIONS

<u>Dnumber</u>	<u>Dlocation</u>
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PROJECT

Pname	<u>Pnumber</u>	Plocation	Dnum
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WORKS_ON

<u>Essn</u>	<u>Pno</u>	Hours
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DEPENDENT

<u>Essn</u>	<u>Dependent_name</u>	Sex	Bdate	Relationship
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ER-to-relational Mapping

STEP1 - Mapping of Regular Entity Types

- A relation R includes all the **simple attributes**
- Choose one of the key attribute as the **PK**
- EMPLOYEE, DEPARTMENT, PROJECT

EMPLOYEE

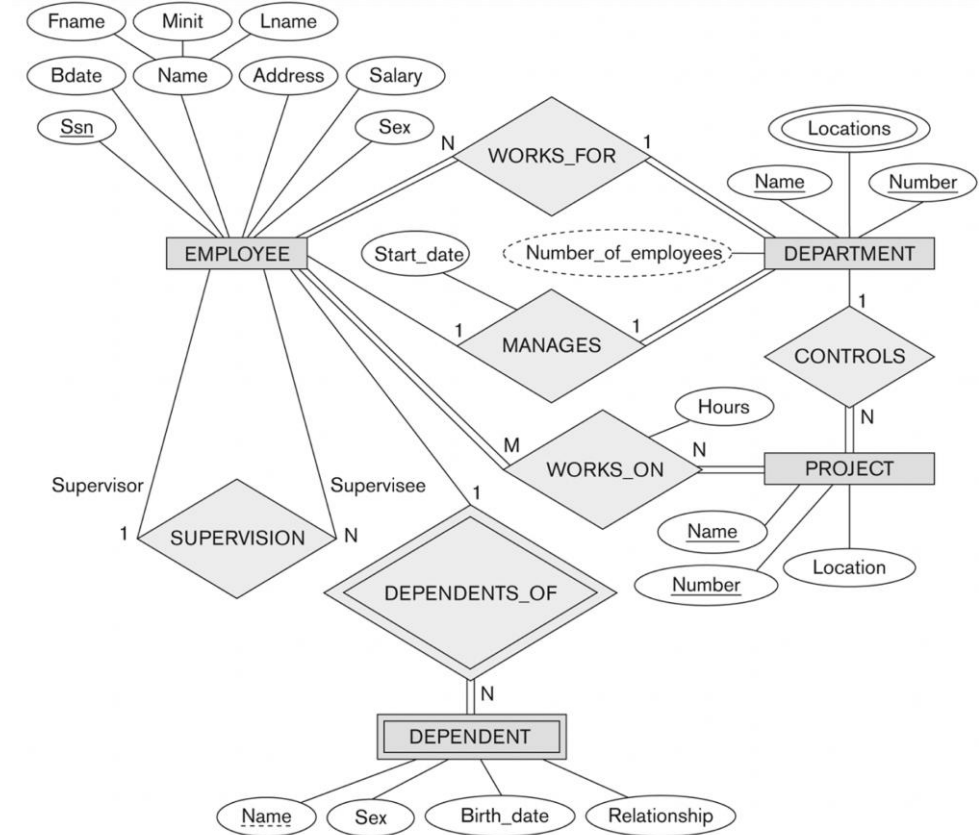
Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary
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DEPARTMENT

Dname	<u>Dnumber</u>
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PROJECT

Pname	<u>Pnumber</u>	Plocation
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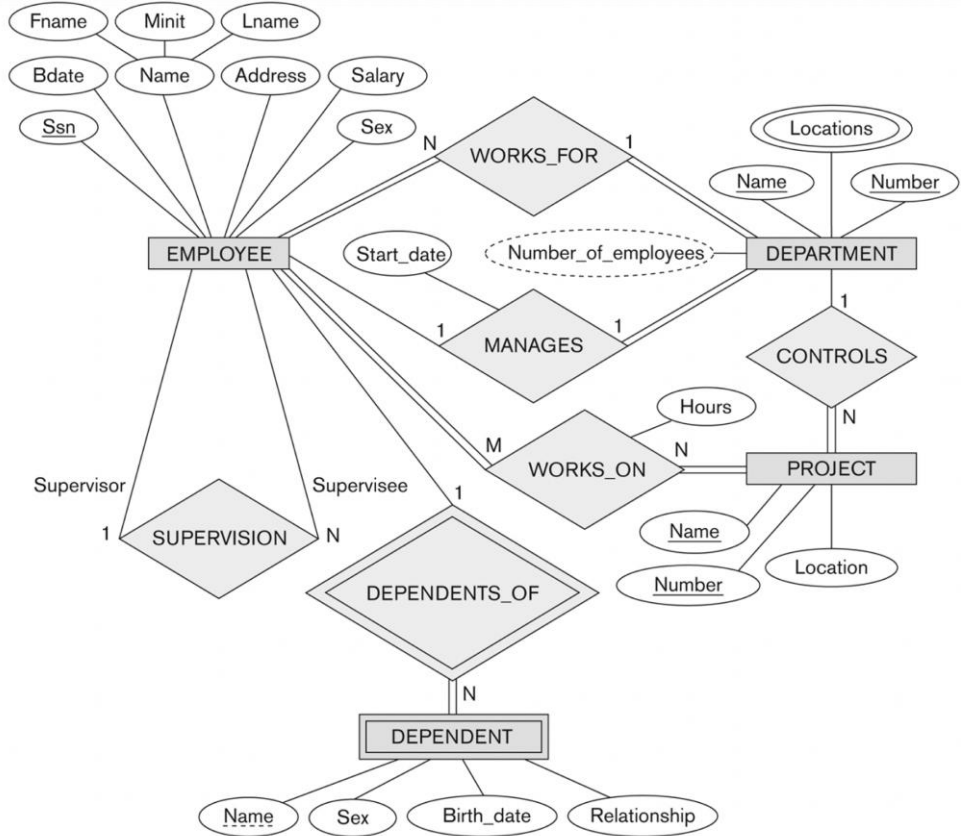
ER-to-relational Mapping

STEP2 - Mapping of Weak Entity Types

- A relation R includes all the **simple attributes** of the weak entity type
- Include the PK attribute(s) of the relation(s) that correspond to the owner entity type(s) as **FK** attributes
- The **PK** of this relation is the combination of the PK(s) of the owner(s) & the partial key of the weak entity type (if any)
- **DEPENDENT**

DEPENDENT

<u>Essn</u>	<u>Dependent name</u>	Sex	Bdate	Relationship
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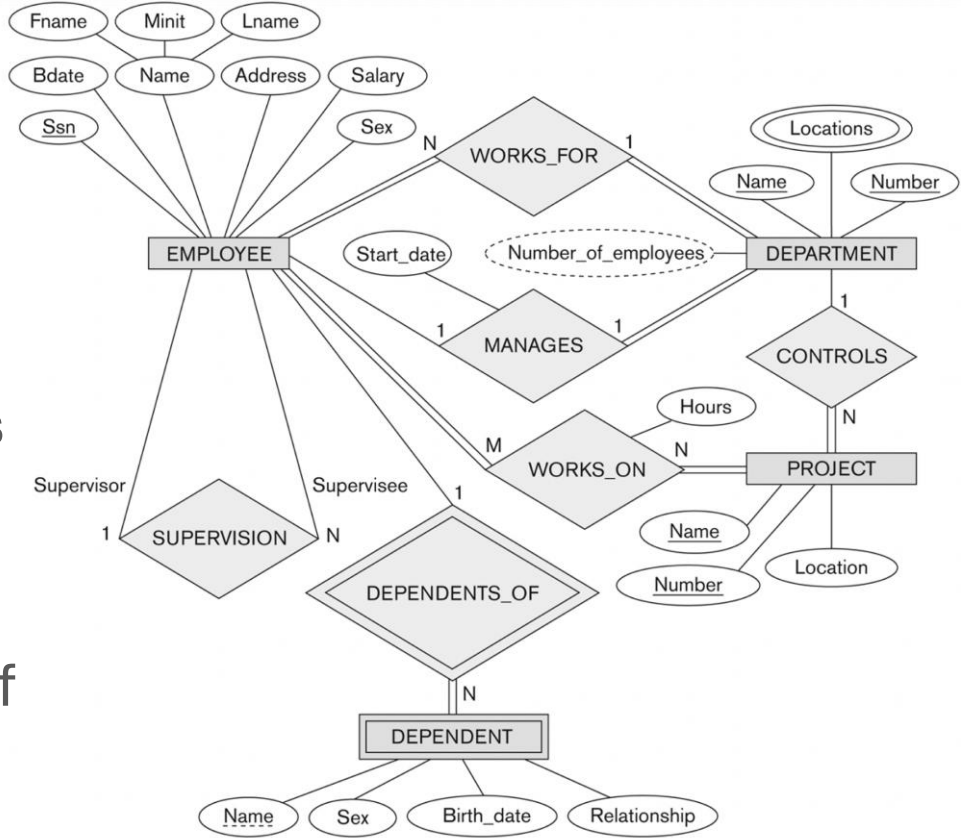
ER-to-relational Mapping

STEP3 - Mapping of Binary 1:1 Relationship Types

- Foreign key approach
 - Choose one of the relations S
 - Include the PK of another relation T as an **FK** of S (better choose an entity type with total participation)
 - Include all the **simple attributes** of the 1:1 relationship type as attribute of S
 - MANAGES

DEPARTMENT

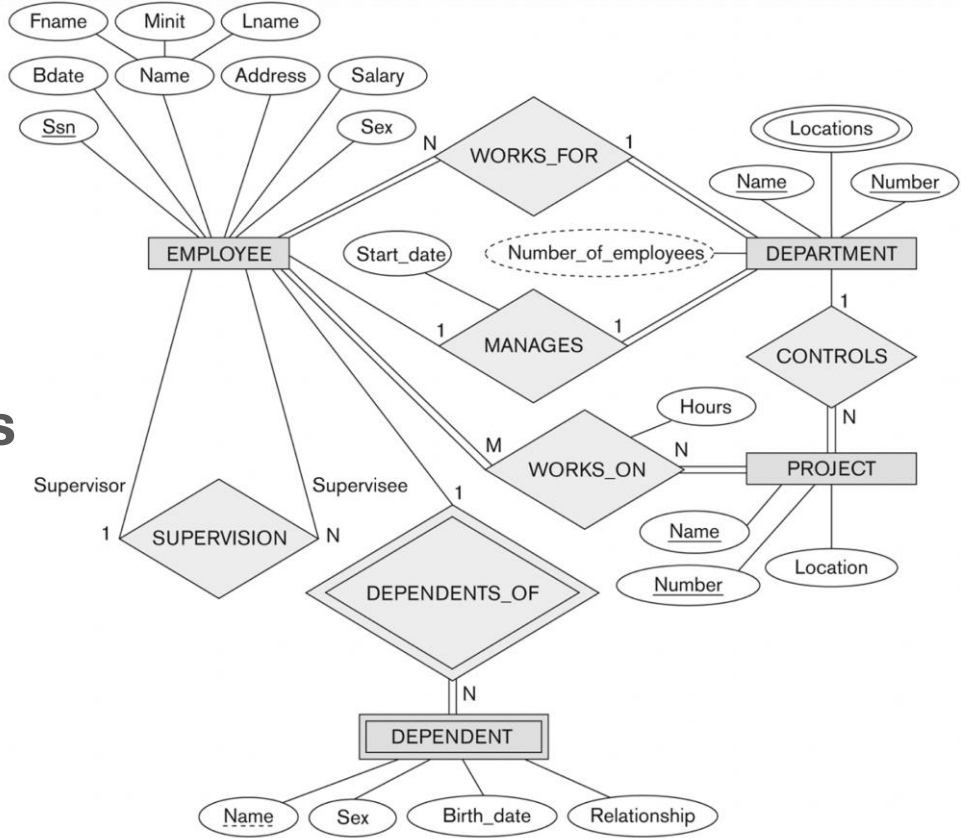
Dname	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
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ER-to-relational Mapping

STEP4 - Mapping of Binary 1:N Relationship Types

- Choose the relations S at the **N-side**
- Include the PK of another relation T as an **FK** of S
- Include all the **simple attributes** of the 1:N relationship type as attribute of S
- SUPERVISION, WORKS_FOR, CONTROLS



EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
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PROJECT

Pname	<u>Pnumber</u>	Plocation	Dnum
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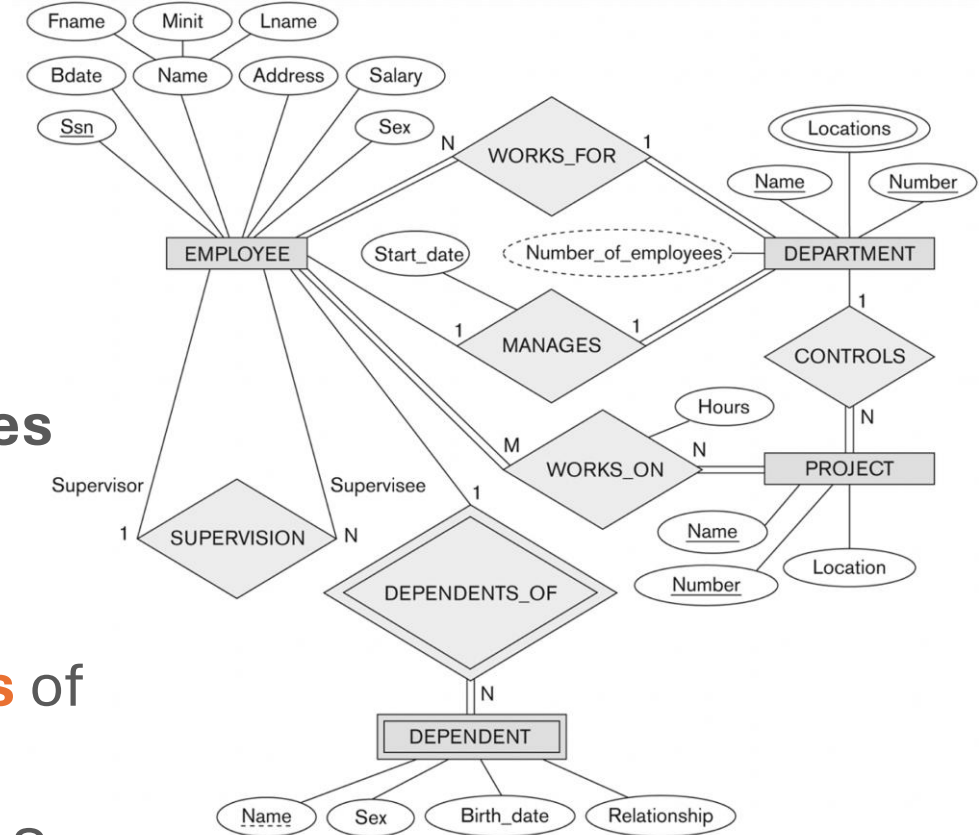
ER-to-relational Mapping

STEP5 - Mapping of Binary M:N Relationship Types

- Create a **new relation** S to represent the relationship
- Include the PK of participating entity types as **FKs** of S
- The combinations of such FKs will form the **PK** of S
- Include all the **simple attributes** of the M:N relationship type as attribute of S
- WORKS_ON

WORKS_ON

<u>Essn</u>	<u>Pno</u>	Hours
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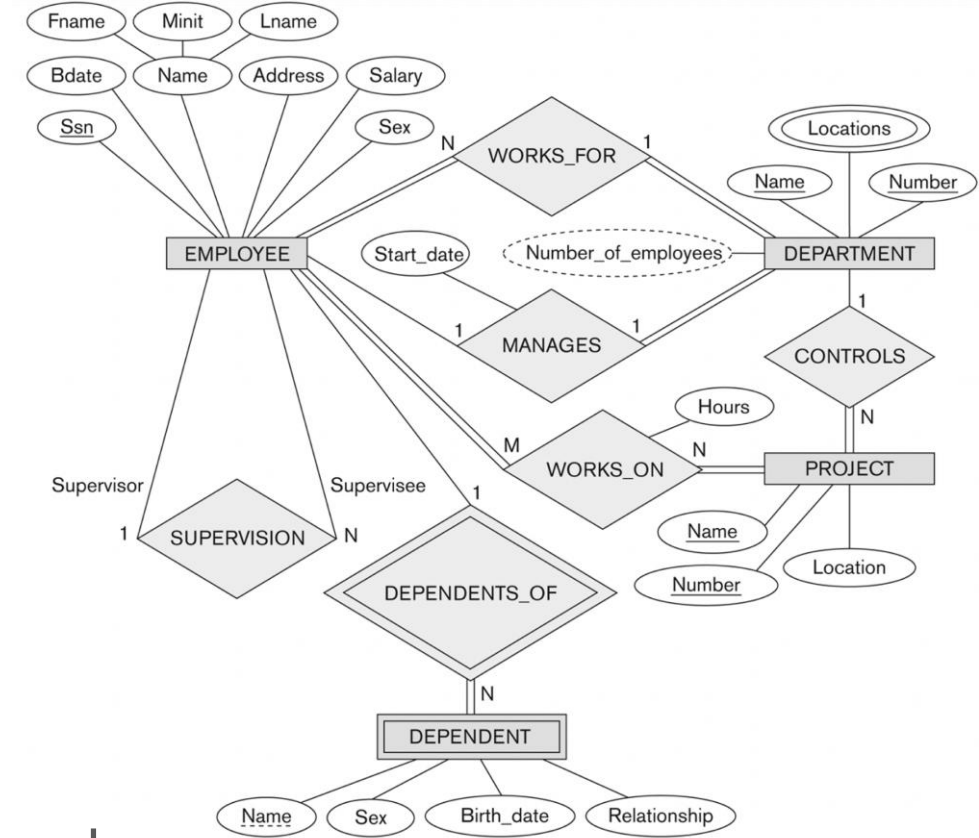
ER-to-relational Mapping

STEP6 - Mapping of Multivalued Attributes

- For each multivalued attribute A, create a **new relation R**
- Include an **attribute corresponding** to A
- Include the PK of the relation that represents the entity/relationship types (that has A as a multivalued attribute) as an **FK** of R
- The **PK** of R is the combination of A and FK
- DEPT_LOCATIONS

DEPT_LOCATIONS

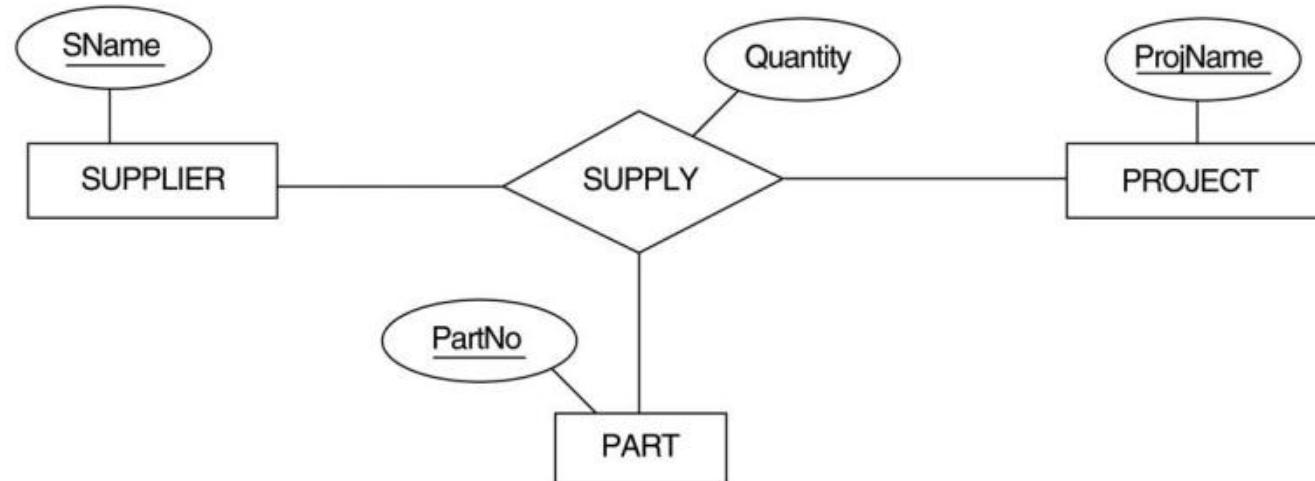
<u>Dnumber</u>	<u>Dlocation</u>
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ER-to-relational Mapping

More mapping

- Mapping of N-ary relationship types
- Mapping specialization or generalization
- Mapping of shared subclasses
- Mapping of categories (union types)



ER-to-relational Mapping

Summary

ER Model	Relational Model
Entity type	Entity relation
1:1 or 1:N relationship type	FK (or relationship relation)
M:N relationship type	Relationship relation and two foreign keys
n-ary relationship type	Relationship relation and n foreign keys
Simple attribute	Attribute
Composite attribute	Set of simple component attributes
Multivalued attribute	Relation and foreign key
Value set	Domain
Key attribute	Primary (or secondary) key

Modification Anomalies

Insertion

EMP_DEPT					Redundancy	
Ename	<u>Ssn</u>	Bdate	Address	Dnumber	Dname	Dmgr_ssn
Smith, John B.	123456789	1965-01-09	731 Fondren, Houston, TX	5	Research	333445555
Wong, Franklin T.	333445555	1955-12-08	638 Voss, Houston, TX	5	Research	333445555
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring, TX	4	Administration	987654321
Wallace, Jennifer S.	987654321	1941-06-20	291 Berry, Bellaire, TX	4	Administration	987654321
Narayan, Ramesh K.	666884444	1962-09-15	975 FireOak, Humble, TX	5	Research	333445555
English, Joyce A.	453453453	1972-07-31	5631 Rice, Houston, TX	5	Research	333445555
Jabbar, Ahmad V.	987987987	1969-03-29	980 Dallas, Houston, TX	4	Administration	987654321
Borg, James E.	888665555	1937-11-10	450 Stone, Houston, TX	1	Headquarters	888665555

- Insert a new employee into EMP_DEPT -> enter all attribute values correctly so that they are **consistent** with the corresponding values in other tuples in EMP_DEPT.
- It is difficult to insert a **new department that has no employee** as yet in the EMP_DEPT. We can place NULL in the attributes of employee but this violates the entity integrity constraint (PK is null).

Modification Anomalies

Deletion

EMP_DEPT					Redundancy	
Ename	<u>Ssn</u>	Bdate	Address	Dnumber	Dname	Dmgr_ssn
Smith, John B.	123456789	1965-01-09	731 Fondren, Houston, TX	5	Research	333445555
Wong, Franklin T.	333445555	1955-12-08	638 Voss, Houston, TX	5	Research	333445555
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring, TX	4	Administration	987654321
Wallace, Jennifer S.	987654321	1941-06-20	291 Berry, Bellaire, TX	4	Administration	987654321
Narayan, Ramesh K.	666884444	1962-09-15	975 FireOak, Humble, TX	5	Research	333445555
English, Joyce A.	453453453	1972-07-31	5631 Rice, Houston, TX	5	Research	333445555
Jabbar, Ahmad V.	987987987	1969-03-29	980 Dallas, Houston, TX	4	Administration	987654321
Borg, James E.	888665555	1937-11-10	450 Stone, Houston, TX	1	Headquarters	888665555

- Delete an employee tuple that happens to represent the last employee working for particular department -> **Lost the information** of that department

Modification Anomalies

Modification (update)

EMP_DEPT					Redundancy	
Ename	<u>Ssn</u>	Bdate	Address	Dnumber	Dname	Dmgr_ssn
Smith, John B.	123456789	1965-01-09	731 Fondren, Houston, TX	5	Research	333445555
Wong, Franklin T.	333445555	1955-12-08	638 Voss, Houston, TX	5	Research	333445555
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring, TX	4	Administration	987654321
Wallace, Jennifer S.	987654321	1941-06-20	291 Berry, Bellaire, TX	4	Administration	987654321
Narayan, Ramesh K.	666884444	1962-09-15	975 FireOak, Humble, TX	5	Research	333445555
English, Joyce A.	453453453	1972-07-31	5631 Rice, Houston, TX	5	Research	333445555
Jabbar, Ahmad V.	987987987	1969-03-29	980 Dallas, Houston, TX	4	Administration	987654321
Borg, James E.	888665555	1937-11-10	450 Stone, Houston, TX	1	Headquarters	888665555

- Update the manager of department 5 -> Update **all the tuples** of all employees who work in department 5

Functional Dependency

- A functional dependency (FD) occurs when the value of one (set of) attribute(s) determines the value of a second (set of) attribute(s)

StudentID $\rightarrow$ StudentName

StudentID $\rightarrow$ {DormName, DormRoom, Fee}

- The attribute on the left side is called the **determinant**.
- A composite determinant is a determinant of an FD that consists of more than one attribute.

Functional Dependency

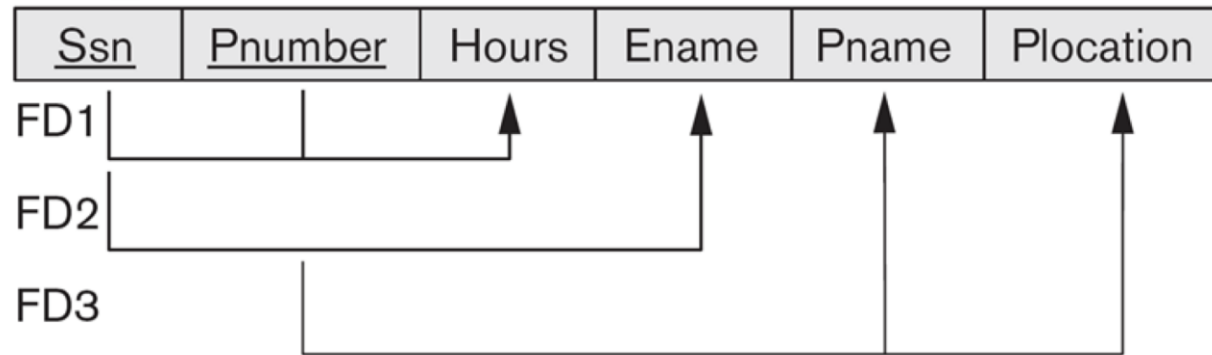
- FDs may be based on equations

ExtendedPrice = Quantity x UnitPrice
 $\{Quantity, UnitPrice\} \rightarrow ExtendedPrice$

- FDs are not equations.
- FD Rules:
 - *If $A \rightarrow (B, C)$, then $A \rightarrow B$ and $A \rightarrow C$.*
 - *If $(A, B) \rightarrow C$, then neither A nor B determines C by itself.*

Functional Dependency

EMP_PROJ



FD1: {Ssn, Pnumber} -> Hours

FD2: Ssn -> Ename

FD3: Pnumber -> {Pname, Plocation}

Normalization of Relations

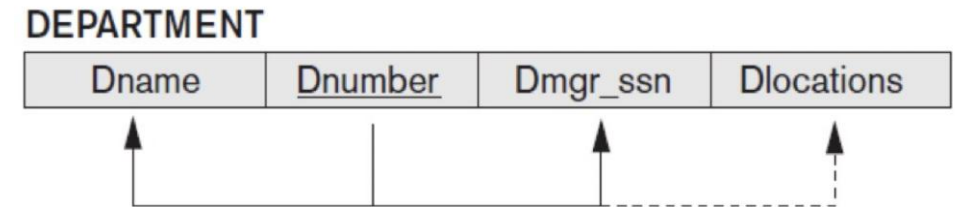
- A process of analyzing the given relation schemas based on their **FDs** and PKs to achieve the **desirable properties**.
 - Minimizing redundancy
 - Minimizing the insertion, deletion, and update anomalies
- Unsatisfactory relation schemas are decomposed into smaller relation schemas that meet the normal form tests.

Normal Forms

- First Normal Form (1NF)
- Second Normal Form (2NF)
- Third Normal Form (3NF)
- Boyce-Codd Normal Form (BCNF)

First Normal Forms

- 1NF *disallows having a set of values or a tuple of values for a single tuple.*
- The only attribute values permitted by 1NF are **single atomic** (or indivisible) **values**.



“Each department can have a number of locations.”

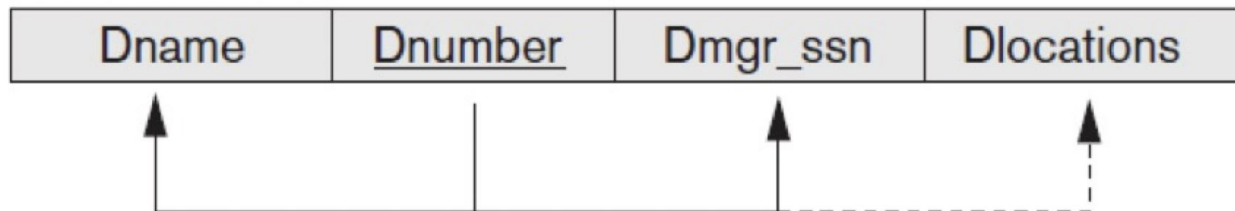
Some tuples can have a set of location values and Dlocations is not FD on the PK Dnumber
The domain of Dlocations contains sets of values and is nonatomic.

First Normal Forms

Achieve 1 NF

1. Create a separate relation for *Dlocations*
DEPT_LOCATIONS: Dnumber, Dlocation
2. Expand the PK to {Dnumber, Dlocation} but introduce redundancy
3. Replace with a certain number of atomic attributes, if maximum number of values is known: Dlocation1, Dlocation2, Dlocation3

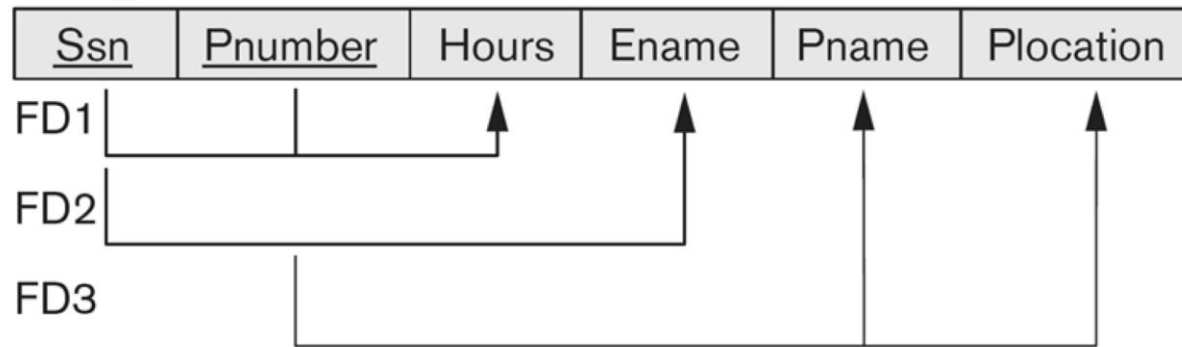
DEPARTMENT



Second Normal Forms

- 2NF is based on the concept of **full functional dependency**.
- Every non-key attribute must be fully functionally dependent on the entire primary key.
- Given $X, Y \rightarrow Z$, the removal of X or Y means that the dependency does not hold anymore.

EMP_PROJ



Full Functional Dependency

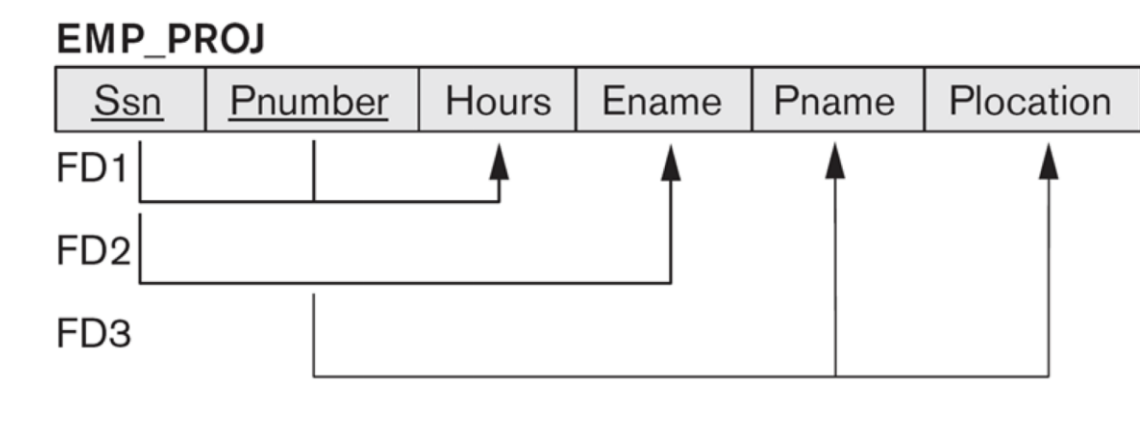
$\{Ssn, Pnumber\} \rightarrow \{Hours, Ename, Pname, Plocation\}$

Partial dependency

$\{Ssn, Pnumber\} \rightarrow Ename$

Second Normal Forms

- All of its non-key attributes are dependent on the whole PK.
- Is this in 1NF/2NF?



FD2: Non-key attribute Ename violates 2NF

FD3: Non-key attributes Ename, Pname, Plocation partially dependent of the PK {Ssn, Pnumber}

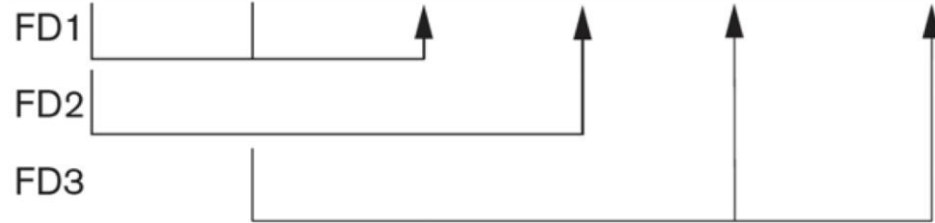
Second Normal Forms

Achieve 2NF

- 2NF normalized into a number of 2NF relations in which *non-key attributes are associated only with the PK(s)*.

EMP_PROJ

<u>Ssn</u>	<u>Pnumber</u>	Hours	Ename	Pname	Plocation
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2NF Normalization

EP1

<u>Ssn</u>	<u>Pnumber</u>	Hours
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EP2

<u>Ssn</u>	Ename
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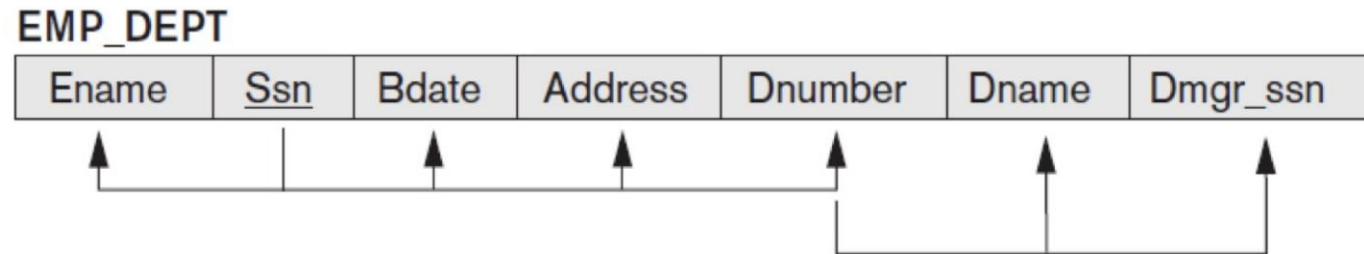
EP3

<u>Pnumber</u>	Pname	Plocation
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Third Normal Forms

- 3NF is based on the concept of **transitive dependency**.
- 3NF is in 2NF & has no determinant except the PK(s)
- Non-key attributes must not depend on other non-key attributes.
- Is this a 2NF/3NF?



Ssn -> Dmgr_ssn is transitive through Dnumber

- Ssn -> Dnumber & Dnumber -> Dmgr_ssn

- Dnumber is not a key or a subset of the key in EMP_DEPT

Third Normal Forms

Achieve 3NF

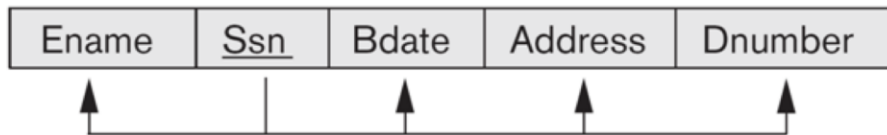
- Normalized into a number of 3NF relations

EMP_DEPT

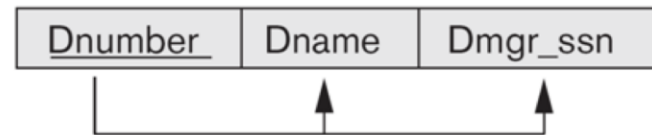


3NF Normalization

ED1



ED2



Normal Forms Based on PK

- Summary

Normal Form	Test	Remedy (Normalization)
First (1NF)	Relation should have no multivalued attributes or nested relations.	Form new relations for each multivalued attribute or nested relation.
Second (2NF)	For relations where primary key contains multiple attributes, no nonkey attribute should be functionally dependent on a part of the primary key.	Decompose and set up a new relation for each partial key with its dependent attribute(s). Make sure to keep a relation with the original primary key and any attributes that are fully functionally dependent on it.
Third (3NF)	Relation should not have a nonkey attribute functionally determined by another nonkey attribute (or by a set of nonkey attributes). That is, there should be no transitive dependency of a nonkey attribute on the primary key.	Decompose and set up a relation that includes the nonkey attribute(s) that functionally determine(s) other nonkey attribute(s).

Boyce-Codd Normal Forms

- BCNF is in 3NF.
- Every determinant is a candidate key.

TEACH

Student	Course	Instructor
Narayan	Database	Mark
Smith	Database	Navathe
Smith	Operating Systems	Ammar
Smith	Theory	Schulman
Wallace	Database	Mark
Wallace	Operating Systems	Ahamad
Wong	Database	Omiecinski
Zelaya	Database	Navathe
Narayan	Operating Systems	Ammar

For every functional dependency ($X \rightarrow Y$), the attribute set X must be a candidate key.

FD1: {Student, Course} $\rightarrow$ Instructor

(Assume that each course has one instructor.)

FD2: Instructor $\rightarrow$ Course

(Assume that each instructor teaches one course.)

> Instructor is not a candidate key.

> The relation is in 3NF but not BCNF.

Surrogate Key

- A synthetic primary key (usually created by DBMS - AUTO_INCREMENT)
- No business or contextual meaning
- Natural Key: a key based on existing data

person_id	name	surname
P0001	Jones	Miler
P0002	Smith	Walker
P0003	William	McQueen

Wrap up

- Create ERD from real-world examples.
- ER-to-relational mapping – 6-step examples
- Modification anomalies: insertion, deletion, and update
- Functional dependencies
- Normalization: 1NF, 2NF, 3NF, and BCNF
- Surrogate Key